

Level 2 — 5 marks/question

L2-Q1. (M) Frequency

P

CODE

```
a=list(map(int,input().split()))
x=max(set(a),key=a.count)
print(x,a.count(x))
```

INPUT

1 2 8 3 2 2 5 1

OUTPUT

2 4

L2-Q2. (M) Starting-Ending

P

CODE

```
a=list(map(int,input().split()))
b=list(map(int,input().split()))
c=list(map(int,input().split()))
target=int(input())
start=-1
end=-1
for i in range(len(a)):
    if a[i]==target:
        if start==-1:
            start=i
            end=i
print(start,end)
```

INPUT

```
5 7 7 8 8 10 8
8
5 7 7 8 8 10
6
[]
0
```

OUTPUT

-1 -1

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L2-Q3. (M) Replace

P

CODE

```
s1="hai welcome to teach coding.it is fun to teach!!!"  
s2="learn"  
s3="teach"  
result=s1.replace(s2,s3)  
print(result)
```

OUTPUT

```
hai welcome to teach coding.it is fun to teach!!!
```

L2-Q4. (M) Lucky

P

CODE

```
n=int(input())  
print("lucky number" if n%9 == 0 else "not lucky")
```

INPUT

```
1415
```

OUTPUT

```
not lucky
```

Student 2 **Chintha Abhilash Reddy** 192525462RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Target**

P

CODE

```
a = list(map(int,input().split()))
target = int(input())
for i in range(len(a)):
    for j in range(i+1,len(a)):
        if a[i] + a[j] == target:
            print(a[i],a[j])
```

INPUT

```
2 4 6 8 10
12
```

OUTPUT

```
2 10
4 8
```

L2-Q2. (M) Frequency

P

CODE

```
a = [1,2,8,3,2,2,2,5,1]
f = {x: a.count(x) for x in set(a)}
print(f)
print(max(f.values()))
```

OUTPUT

```
{1: 2, 2: 4, 3: 1, 5: 1, 8: 1}
4
```

L2-Q3. (M) Function Vowels

P

CODE

```
s1 = input()
s2 = input()
s3 = " "
for i in range(len(s1)):
    s3+= s1[i]
    if i<len(s2):
        s3+= s2[i]
print(s3)
```

INPUT

```
Welcome
homely
```

OUTPUT

```
Wheolmceolmye
```

L2-Q4. (M) Lucky

P

CODE

```
n = int(input())
if n == 1412:
    print("yes")
else:
    print("no")
```

INPUT

```
1412
```

OUTPUT

```
yes
```


Student 3 **DANIMIZA V** 192421363RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Minimum**

P

CODE

```
a = [0,1,4,4,5,6,7]
n = 4
for i in range(n):
    x = a.pop()
    x = a.insert(0,x)
print(a)
print(min(a))
```

OUTPUT

```
[4, 5, 6, 7, 0, 1, 4]
0
```

L2-Q2. (M) Starting-Ending

P

CODE

```
nums = [5,7,7,8,8,10]
target = 8
found = False
for i in range(len(nums)):
    for j in range(i+1, len(nums)):
        if nums[i] + nums[j] == target:
            print(nums[i], nums[j])
            found = True
```

OUTPUT

L2-Q3. (M) Isomorphic

P

CODE

```
def isomorphic(s,t):
    if len(s) != len(t):
        return False
    d = {}
    for i in range(len(s)):
        if s[i] in d and d[s[i]] != t[i]:
            return False
        d[s[i]] = t[i]
    return len(set(d.values())) == len(d)
print(isomorphic("egg", "add"))
```

OUTPUT

True

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L2-Q4. (M) Lucky

P

CODE

```
n = 1412
while n ≥ 10:
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    n = s
if n == 7:
    print("lucky")
else:
    print("not lucky")
```

OUTPUT

not lucky

Student 4 **DEVALLA GURU RAMYA SREE** 192420061RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Frequency

P

CODE

```
a = eval(input())

freq = {}

for x in a:
    if x in freq:
        freq[x] += 1
    else:
        freq[x] = 1
    print("Element | Frequency")
    print("—————")

    for x in freq:
        print("x,          |",freq[x])

    m = max(freq.values())
    for x in freq:
        if freq in x == [m]:
            print("output", x, "&", m)
```

OUTPUT

Security Error: Disallowed code pattern detected.

L2-Q2. (M) 2D Multiplication

P

CODE

```
a = [[1, 2], [5,3]]
b = [[2,3], [4,1]]
c = [[sum(a[i][k] * b[k][j] for k in range (2)) for j in range (2)]for i in range(2)]

for row in c:
    print(*row)
```

INPUT

```
1 2
5 3
2 3
4 1
```

OUTPUT

```
10 5
22 18
```

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L2-Q3. (M) Words-Lines

P

CODE

```
import sys
lines = sys.stdin.read().splitlines()
print("No. of words in lines =",len(lines[0].split()))
print("No. of lines in files =",len(lines))
```

INPUT

"Machine learning (ML) is a kind of AI (artificial intelligence) that lets software applications become more precise at anticipating results without being specifically programmed to do so. Machine learning algorithms utilize historical data to anticipate/predict the latest output value S."

OUTPUT

No. of words in lines = 13
No. of lines in files = 4

L2-Q4. (M) Lucky

P

CODE

```
def rsum(n):
    if n == 0:
        return 0
    return n % 10 + rsum(n // 10)

def rprod(n):
    if n < 10:
        return n
    return n % 10 * rprod(n // 10)

n = (int(input()))

if rsum(n) == rprod(n):
    print("YES")
else:
    print("NO")
```

INPUT

1412

OUTPUT

YES

Student 5 **Deveswar A** 192512152RC

3 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Sort

P

CODE

```
n = list(map(int,input().split()))
c = sorted(n)
for i in c:
    if len(c)//2==1:
        print (c[i])

    else:
        print ("Middle element :",c[(len(c)//2)-1])
        break
```

INPUT

90 -90 80 66 78 10

OUTPUT

Middle element : 66

L2-Q2. (M) Target

P

CODE

```
n = list(map(int,input().split()))
t = 10
for i in range(0,len(n)):
    for j in range (i+1,len(n)-1):
        c = n[i]+n[j]
        if t == c:
            print (n[i],n[j])
```

INPUT

2 8 4 7 9

OUTPUT

2 8

L2-Q3. (M) Palindromic

— Not Submitted —

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L2-Q4. (M) Perfect Square

P

CODE

```
import math
n = int (input())
s = 0
count = 0
for i in range(1,n+1):
    if ((math.sqrt(i))**2)==i:
        count+=1
print (count)
```

INPUT

4

OUTPUT

2

Student 6 **H Harshavardhan** 192511171RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Starting-Ending

P

CODE

```
a=list(map(int,input().split()))
b=int(input())
s=-1
e=-1
for i in range(len(a)):
    if a[i]==b:
        if s==-1:
            s=i
        e=i
print([s,e])
```

INPUT

```
5 7 7 8 8 10
8
```

OUTPUT

```
[3, 4]
```

L2-Q2. (M) Minimum

P

CODE

```
n=list(map(int,input().split()))
a=int(input())
for i in range(a):
    n=[n[-1]]+n[:-1]
print(n)
print(min(n))
```

INPUT

```
0 1 4 4 5 6 7
7
```

OUTPUT

```
[0, 1, 4, 4, 5, 6, 7]
0
```

L2-Q3. (M) Valid String

P

CODE

```
str=input()
if str=='()' or str=='[]' or str=='{}':
    print("true")
elif str=="(){}" or str=="[]{}" or str=="()[]" or str=="[](){}":
    print("true")
else:
    print("false")
```

INPUT

```
()
```

OUTPUT

```
true
```

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L2-Q4. (M) Perfect Square

P

CODE

```
a=int(input())
b=0
for i in range(1,a+1):
    x=i*(1/2)
    if x==int(x):
        b+=1
print(b)
```

INPUT

12

OUTPUT

3

Student 7 **Jayadeep J** 192511304RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Search

P

CODE

```
b=False
a=[]
for i in range(5):
    x=list(map(int,input().split()))
    a.append(x)
t=int(input())
for i in range(len(a)):
    for j in range(len(a[i])):
        if int(a[i][j])==t:
            b=True
            break
print(b)
```

INPUT

```
1 4 7 11 15
2 5 8 12 19
3 6 9 16 22
10 13 14 17 24
18 21 23 26 30
5
```

OUTPUT

True

L2-Q2. (M) Target

P

CODE

```
a=[int(i) for i in input().split(" ")]
b=int(input())
for i in range(len(a)):
    for j in range(i,len(a)):
        if a[i]+a[j]==b:
            x=i+1,j+1
            x=list(x)
            print(x)
```

INPUT

```
1 2 3 4 5 6 7 8 9
12
```

OUTPUT

```
[3, 9]
[4, 8]
[5, 7]
[6, 6]
```

L2-Q3. (M) Palindromic

P

CODE

```
a=input()
for i in range(len(a)):
    for j in range(i,len(a)):
        x=a[i:j]
        if x==x[::-1] and len(x)>1:
            print(x)
```

INPUT

cbbd

OUTPUT

bb

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L2-Q4. (M) Perfect Square

P

CODE

```
a=int(input())
c=0
for i in range(1,a+1):
    x=i*(1/2)
    if x==int(x):
        c+=1
print(c)
```

INPUT

13

OUTPUT

3

Student 8 **Kampati Pavan Kumar** 192524333RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Target**

P

CODE

```
a=list(map(int,input().split()))
n=int(input())
length=len(a)
b=[]
for i in range(length):
    for j in range(i+1,length):
        if n==a[i]+a[j]:
            b.append([a[i],a[j]])
            print(b)
```

INPUT

```
1 2 3 4 5 6 7 8 9
12
```

OUTPUT

```
[[3, 9]]
[[3, 9], [4, 8]]
[[3, 9], [4, 8], [5, 7]]
```

L2-Q2. (M) Peak element

P

CODE

```
a=list(map(int,input().split()))
for i in a:
    if a[i] > a[i-1] and a[i] > a[i+1]:
        print(i)
```

INPUT

```
1 2 3 1
```

OUTPUT

```
2
```

L2-Q3. (M) Replace

P

CODE

```
s1=input()
s2=input()
s3=input()
print(s1.replace(s2,s3))
```

INPUT

```
Hai welcome to learn coding. It is fun to learn!!
welcome
"
```

OUTPUT

```
Hai " " to learn coding. It is fun to learn!!
```

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L2-Q4. (M) Perfect Square

P

CODE

```
def square(n):  
    dp=[0]+[n]*n  
    for i in range(1,n+1):  
        for j in range(1,int(i**0.5)+1):  
            dp[i]=min(dp[i],dp[i-j*j]+1)  
    return dp[n]  
n=int(input())  
print(square(n))
```

INPUT

3

OUTPUT

3

Student 9 **KARTHIGA S** 192319057RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Minimum

P

CODE

```
a=list(map(int,input().split()))
n=int(input())
n=n-1
res=a[n:]+a[:n]
print(res,"if it was rotated 4 times.")
```

INPUT

```
0 1 4 4 5 6 7
4
```

OUTPUT

```
[4, 5, 6, 7, 0, 1, 4] if it was rotated 4 times.
```

L2-Q2. (M) Peak element

P

CODE

```
a=list(map(int,input().split()))
for i in range(1,len(a)-1):
    if a[i]>a[i+1] and a[i]>a[i-1]:
        b=a[i]
        for x in a:
            if a[i]==b:
                print(i)
                break
```

INPUT

```
1 2 3 1
```

OUTPUT

```
2
```

L2-Q3. (M) Function Vowels

P

CODE

```
s1=input()
s2=input()
n=len(s2)
s3=""
for i in s1:
    s3+=i
    s3=s3+s2[i]
print("S3:",s3)
c=0
for ch in s3:
    if ch.lower() in "aeiou":
        c+=1
print(f"Vowel count in string S3 is {c}")
```

INPUT

```
Welcome
Homely
```

OUTPUT

```
W H e o l m c e o l m y e a
Vowel count in string S3 is 5
```

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L2-Q4. (M) Perfect Square

P

CODE

```
n=12
dp=[0]+[float('inf')]*n
for i in range(1,n+1):
    j=1
    while j*j≤i:
        dp[i]=min(dp[i],dp[i-j*j]+1)
        j+=1
print(dp[n])
```

INPUT

12

OUTPUT

3

Student 10 **KAVI PREETHA G** 192420040RC

3 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Minimum

P

CODE

```
a = list(map(int,input().split()))
a.sort()
print(min(a))
```

INPUT

4 5 6 7 1 0 4

OUTPUT

0

L2-Q2. (M) Target

— Not Submitted —

L2-Q3. (M) Palindromic

P

CODE

```
s = input()
print(s if s == s[::-1] else "not found")
```

INPUT

bab

OUTPUT

bab

L2-Q4. (M) Lucky

P

CODE

```
n = int(input())
s = 0
p = 1
for x in range(n):
    x = int(x)
    s += x
    p *= x
if s == p:
    print("No")
else:
    print("Yes")
```

INPUT

1412

OUTPUT

Yes

Student 11 **Kolli Venkata Yathish Chandra** 192524323RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Frequency**

P

CODE

```
a=[1,2,8,3,2,2,2,5,1]
```

OUTPUT

2 & 4

L2-Q2. (M) Starting-Ending

P

CODE

```
a=[5,7,7,8,8,10]
target=8

start = -1
end = -1

for i in range(len(a)):
    if a[i] == target:
        if start == -1:
            start = i
        end = i
print(start, end)
```

OUTPUT

3 4

L2-Q3. (M) Palindromic

P

CODE

```
s="babad"
longest=" "

for i in range(len(s)):
    for j in range(i,len(s)):
        sub = s[i:j+1]
        if sub == sub[::-1] and len(sub) > len(longest):
            longest=sub
print(longest)
```

OUTPUT

bab

L2-Q4. (M) Lucky

P

CODE

```
n=int(input())

if n%2 == 1:
    print("no")
else:
    print("yes")
```

INPUT

1412

OUTPUT

yes

Student 12 **Kondam Bhaskar Tharunika** 192525276RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Minimum**

P

CODE

```
def findMin(nums):
    left = 0
    right = len(nums) - 1
    while left < right:
        mid = (left+right) // 2
        if nums[mid]>nums[right]:
            left = mid + 1
        elif nums[mid]<nums[right]:
            right = mid
        else:
            right -= 1
    return nums[left]
nums = [0,1,4,4,5,6,7]
print(findMin([4,5,6,7,0,1,4])) #0
print(findMin([0,1,4,4,5,6,7])) #0
```

OUTPUT

```
0
0
```

L2-Q2. (M) Sort

P

CODE

```
arr = [90,-90,80,66,78,10]
arr.sort()
print("Sorted array:",arr)
middle = arr[len(arr) // 2]
print("Middle element:",middle)
```

OUTPUT

```
Sorted array: [-90, 10, 66, 78, 80, 90]
Middle element: 78
```

L2-Q3. (M) Replace

P

CODE

```
s1 = input()
s2 = input()
s3 = input()
print(s1.replace(s2,s3))
```

INPUT

```
s1: Hai welcome to learn coding. It is fun to learn!!!
s2: Code
s3: Teach
```

OUTPUT

```
s1: Hai welcome to learn coding. It is fun to learn!!!
```

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L2-Q4. (M) Lucky

P

CODE

```
int(input())  
print("yes")  
print("1.-1412")  
print("2.14.12")  
print("3.1415")
```

INPUT

1412

OUTPUT

```
yes  
1.-1412  
2.14.12  
3.1415
```

Student 13 **Kondapalli Pujitha** 192524188RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Starting-Ending**

P

CODE

```
arr = list(map(int,input().split()))
target = int(input())
if target in arr:
    print ([arr.index(target),len(arr)-1-arr[::-1].index(target)])
else:
    print([-1,-1])
```

INPUT

```
5 7 7 8 8 10
8
```

OUTPUT

```
[3, 4]
```

L2-Q2. (M) Sort

P

CODE

```
arr= list(map(int,input().split()))
arr.sort()
print(*arr)
print("middle element:",arr[(len(arr)-1)//2])
```

INPUT

```
90 -90 80 66 78 10
```

OUTPUT

```
-90 10 66 78 80 90
middle element: 66
```

L2-Q3. (M) Words-Lines

P

CODE

```
import sys
lines = sys.stdin.read().splitlines()
print("No. of words in lines =",len(lines[0].split()))
print("No. of lines in file =",len(lines))
```

INPUT

```
"Machine learning(ML) is a kind of AI (artificial intelligence) that lets software applications become more precise at anticipating
results without being specifically programmed.
```

OUTPUT

```
No. of words in lines = 23
No. of lines in file = 1
```

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L2-Q4. (M) Perfect Square

P

CODE

```
import math
n = int(input())
dp = [0]+[n]*n
for i in range(1,n+1):
    for j in range(1,math.isqrt(i)+1):
        dp[i]=min(dp[i],dp[i-j*j]+ 1)
        j+=1
print (dp[n])
```

INPUT

12

OUTPUT

3

Student 14 **KOTA PAVITHRA** 192323011RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Target

P

CODE

```
a=list(map(int,input().split()))
target=int(input())
for i in range(len(a)):
    for j in range(i+1,len(a)):
        if a[i]+a[j]==target:
            print(a[i+1],a[j+1])
```

INPUT

```
1 2 3 4 5
3
```

OUTPUT

```
2 3
```

L2-Q2. (M) Intersection

P

CODE

```
a=[1,2,2,1]
b=[2,2]
intersection=[]
for num in a:
    if num in b:
        intersection.append(num)
        b.remove(num)
print(intersection)
```

OUTPUT

```
[2, 2]
```

L2-Q3. (M) Valid String

P

CODE

```
s=input()
stack=[]
valid=True
for ch in s:
    if ch in "[]{()}":
        print("true")
    else:
        print("false")
```

INPUT

```
()
```

OUTPUT

```
true
true
false
```

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L2-Q4. (M) Perfect Square

P

CODE

```
n=int(input("Enter n:"))
dp=[0] + [n] * n
for i in range(1,n+1):
    j=1
    while j * j ≤ i:
        dp[i]=min(dp[i], dp[i-j*j]+1)
        j+=1
print("Least perfect square:",dp[n])
```

INPUT

12

OUTPUT

Enter n:Least perfect square: 3

Student 15 **Madireddy Gowtham Kumar Reddy** 192572066RC

3 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) 2D Multiplication**

P

CODE

```
a = [[1,2],[5,3]]
b = [[2,3],[4,1]]
c = [[0,0],[0,0]]
for i in range(2):
    for j in range(2):
        for k in range(2):
            c[i][j]+=a[i][k]*b[k][j]
print(c)
```

OUTPUT

[[10, 5], [22, 18]]

L2-Q2. (M) Frequency

P

CODE

```
arr = list(map(int,input().split()))
max_count = 0
element = None
for x in arr:
    count = arr.count(x)
    if count > max_count:
        max_count = count
        element = x
print("Element:",element)
print("Frequency:",max_count)
```

INPUT

1 2 8 3 2 2 2 5 1

OUTPUT

Element: 2
Frequency: 4**L2-Q3. (M) Function Vowels**

— Not Submitted —

L2-Q4. (M) Lucky

P

CODE

```
n = 1412
if n%2==1:
    print("not lucky")
else:
    print("lucky")
```

OUTPUT

lucky

Student 16 **MALAVIKAH V** 192419061RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Starting-Ending

P

CODE

```
import bisect
def searchrange(arr,t):
    left=bisect.bisect_left(arr,t)
    if left == len(arr) or arr[left]!=t:
        return [-1,-1]
    right = bisect.bisect_right(arr,t)-1
    return [left, right]
arr=[5,7,7,8,8,10]
t=8
print(searchrange(arr,t))
```

OUTPUT

```
File "C:\Windows\TEMP\sandbox_192419061RC_6a7d61253f4630.04519043\main.py", line 6
    right = bisect.bisect_right(arr,t)-1
                                ^
```

```
IndentationError: unindent does not match any outer indentation level
```

L2-Q2. (M) Search

P

CODE

```
s=input()
t=int(input())
n=list(map(int,s.replace('[','').replace(' ','').split(',')))
cols=5
mat=[n[i:i+cols] for i in range(0,len(n),cols)]
i=0
j=cols-1
while i<len(mat) and j>=0:
    if mat[i][j]==t:
        print("True")
        break
    elif mat[i][j]>t:
        j-=1
    else:
        i+=1
else:
    print("False")
```

INPUT

```
[[1,4,7,11,15],[2,5,8,12,19],[3,6,9,16,22],[10,13,14,17,24],[18,21,23,26,30]]
5
```

OUTPUT

```
True
```


L2-Q3. (M) Valid String

p

CODE

```
s=input()
stack=[]
pairs={'(': ')': '[', ']': '{', '}'}
for c in s:
    if c in ")]}":
        stack.append(c)

    elif c in "[{(":
        if not stack or stack[-1] != pairs[c]:
            print("False")
            break
else:
    print("True" if not stack else "False")
```

INPUT

S = "()"

OUTPUT

False

L2-Q4. (M) Perfect Square

p

CODE

```
def numsquares(n):
    squares = [i*i for i in range(1, int(n**0.5)+1)]
    queue = [n]
    steps = 0

    while queue:
        next_queue=[]
        for num in queue:
            if num == 0:
                return steps
            for sq in squares:
                if sq ≤ num:
                    next_queue.append(num - sq)
        queue = list(set(next_queue))
        steps += 1

print(numsquares(12))
```

OUTPUT

3

Student 17 **Mohammed Subahaan H** 192572165RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Frequency

P

CODE

```
n=input().replace("[","")
m=n.replace("]", "")
l=list(map(int,m.split(",")))
print("Element | Frequency")
print("-----")
a=list(set(l))
a.sort()
for i in a:
    print(i,"      |",l.count(i))
c=0
for i in a:
    if(c<l.count(i)):
        c=l.count(i)
        e=i
print(e," & ",c)
```

INPUT

[1,2,8,3,2,2,2,5,1]

OUTPUT

```
Element | Frequency
-----
1       | 2
2       | 4
3       | 1
5       | 1
8       | 1
2 & 4
```

L2-Q2. (M) Repeated

P

CODE

```
n=input().split("nums = ")[1]
seen=[]
dup=0
for i in n:
    if(i not in seen):
        seen.append(i)
    else:
        dup=i
print(dup)
```

INPUT

nums = [1,2,3,4,2]

OUTPUT

2

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L2-Q3. (M) Words-Lines

P

CODE

```
a=input()
l=a.split(". ")
c=0
for i in l:
    c+=len(i.split(" "))
print("No. of words in a line = ",round(c/len(l)))
cl=0
for i in l:
    cl+=1
print("No. of line in a file = ",cl)
```

INPUT

Machine learning (ml) is a kind of AI that lets software application become more precise at anticipating results without being specifically programmed to do so. machine learning algorithm utilize historical data as input to anticipate / predict the latest output values.

OUTPUT

No. of words in a line = 20
No. of line in a file = 2

L2-Q4. (M) Lucky

P

CODE

```
def rSum(n):
    if(n<10):
        return n
    else:
        return n%10+rSum(n//10)

def rProd(v):
    if(v<10):
        return v
    else:
        return v%10*rProd(v//10)

i=int(input())
sum=rSum(i)
prod=rProd(i)
if(sum==prod):
    print("Yes")
else:
    print("NO")
```

INPUT

1412

OUTPUT

Yes

Student 18 **Moniga V** 192521184RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Repeated

P

CODE

```
def findDuplicate(nums):
    slow = nums[0]
    fast = nums[0]
    while True:
        slow = nums[slow]
        fast = nums[nums[fast]]
        if slow == fast:
            break
    slow = nums[0]
    while slow != fast:
        slow = nums[slow]
        fast = nums[fast]
    return slow

nums = [1,3,4,2,2]
print(findDuplicate(nums))
```

INPUT

nums = [1,3,4,2,2]

OUTPUT

2

L2-Q2. (M) Sort

P

CODE

```
arr = [90,-90,80,66,78,10]

arr.sort()
print(arr)

n = len(arr)
middle = arr[n // 2 - 1] if n % 2 == 0 else arr[n // 2]
print("Middle element=", middle)
```

OUTPUT

```
[-90, 10, 66, 78, 80, 90]
Middle element= 66
```

L2-Q3. (M) Replace

P

CODE

```
S1 = "Hai Welcome to learn coding. It is fun to learn!!!"
S2 = "learn"
S3 = "teach"

result = S1.replace(S2, S3)
print(result)
```

OUTPUT

Hai Welcome to teach coding. It is fun to teach!!!

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L2-Q4. (M) Perfect Square

P

CODE

```
def numsquares(n):  
    squares = [i*i for i in range(1, int(n**0.5)+1)]  
    queue = [n]  
    steps = 0  
  
    while queue:  
        next_queue=[]  
        for num in queue:  
            if num == 0:  
                return steps  
            for sq in squares:  
                next_queue.append(num - sq)  
        queue = list(set(next_queue))  
        steps += 1  
  
print(numsquares(12))
```

OUTPUT

3

Student 19 **Mrudhula K** 192524335RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Peak element

P

CODE

```
def peak(a):
    for i in a:
        if a[i]>a[i+1] or a[i]≥a[i-1]:
            return a[i]
a=list(map(int,input().split()))
print(peak(a))
```

INPUT

1 2 3 1

OUTPUT

2

L2-Q2. (M) Sort

P

CODE

```
n=int(input())
arr=list(map(int,input().split()))
a=arr[n//2]
arr.sort()
print(arr)
print("Middle element :", a )
```

INPUT

6
90 -90 80 66 78 10

OUTPUT

[-90, 10, 66, 78, 80, 90]
Middle element : 66

L2-Q3. (M) Number of Ups, Lows

P

CODE

```
upper=0
lower=0
number=0
while True:
    s=input()
    if s=="*":
        break
    for ch in s:
        if ch.islower():
            lower+=1
        if ch.isupper():
            upper+=1
        if ch.isdigit():
            number+=1
print("no of lower case:", lower)
print("no of upper case:", upper)
print("no of numbers:", number)
```

INPUT

W
d
A
G
g
H
*

OUTPUT

no of lower case: 2
no of upper case: 4
no of numbers: 0

L2-Q4. (M) Perfect Square

P

CODE

```
n=int(input())
dp=[0]*(n+1)
for i in range(1,n+1):
    dp[i]=i
    j=1
    while j*j≤i:
        dp[i]=min(dp[i] , dp[i-j*j]+1)
        j+=1
print(dp[n])
```

INPUT

12

OUTPUT

3

Student 20 **Nichenametla Venkata Mani Chandan** 192525392RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Repeated

P

CODE

```
n = [1,3,4,2,2]
for i in range(len(n)):
    for j in range(i+1,len(n)):
        if n[i]==n[j]:
            print(n[i])
            break
```

OUTPUT

2

L2-Q2. (M) Search

P

CODE

```
a=[[1,4,7,11,15],
   [2,5,8,12,19],
   [3,6,9,16,22],
   [10,13,14,17,24],
   [18,21,23,26,30]]
target=5

for i in range(len(a)):
    for j in range(i+1,len(a)):
        if a[i]+a[j]==target:
            print("false")
        else:
            print("true")
```

OUTPUT

```
true
true
true
true
true
true
true
true
true
true
```

L2-Q3. (M) Valid String

P

CODE

```
s = "()[{}]"

if s== "()[{}]":
    print("true")
else:
    print("false")
```

OUTPUT

true

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L2-Q4. (M) Perfect Square

P

CODE

```
n = 1
count = 0

while n>0:
    i = 1
    while(i + 1) * (i + 1) ≤ n:
        i +=1

    n= n-(i*i)
    count +=1
print(count)
```

OUTPUT

1

Student 21 **Nithish S** 192521249RC

3 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Peak element

P

CODE

```
def peak(a):
    for i in range(len(a)):
        if(i==0 or a[i] ≥ a[i-1] and (i==len(a)-1) or a[i] ≥ a[i+1]):
            return a[i]
a=[2,2,3,1]
print(peak(a))
```

OUTPUT

2

L2-Q2. (M) Repeated

P

CODE

```
a=[3,1,3,4,2]
for x in set(a):
    if a.count(x)>1:
        print(x)
```

OUTPUT

3

L2-Q3. (M) Haystack

P

CODE

```
s=input("")
x=input("")
print(s.find(x))
```

INPUT

sadbutsad
sad

OUTPUT

0

L2-Q4. (M) Lucky

— Not Submitted —

Student 22 **P V Karishmika** 192511453RC

3 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Peak element**

P

CODE

```
def peak(a):
    n=len(a)
    for i in range(n):
        if(i==0 or a[i]≥a[i-1]) and (i==n-1 or a[i]≥a[i+1]):
            return a[i]
a=list(map(int,input().split()))
print(peak(a))
```

INPUT

1 2 3 1

OUTPUT

3

L2-Q2. (M) Search

P

CODE

```
m,n=map(int,input().split())
a=[]
for i in range(m):
    a.append(list(map(int,input().split())))
target=int(input())
found=false
for i in range(m):
    for j in range(n):
        if a[i][j]==target:
            found=true
if found:
    print("true")
else:
    print("false")
```

INPUT

```
5
1 2
4 5
7 8
11 12
15 19
```

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192511453RC_6a7d4e4d6a34e1.71748321\main.py", line 1, in <module>
    m,n=map(int,input().split())
    ^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

L2-Q3. (M) Valid String

— Not Submitted —

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L2-Q4. (M) Lucky

P

CODE

```
def lucky(n):  
    s=0  
    while n>0:  
        s+=n%10  
        n//=10  
    return s==8  
n=int(input())  
if lucky(n):  
    print("lucky")  
else:  
    print("not lucky")
```

INPUT

1412

OUTPUT

lucky

Student 23 **PATAN MOHAMMED RIYAZ KHAN** 192572126RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) 2D Multiplication**

P

CODE

```
a=[
    [1,2],
    [5,3]
]
b=[
    [2,3],
    [4,1]
]
c=[
    [0,0],
    [0,0]
]
for i in range(2):
    for j in range(2):
        for k in range(2):
            c[i][j]+=a[i][k]*b[k][j]

for row in c:
    print(row)
```

INPUT

```
1 2 5 3
2 3 4 1
```

OUTPUT

```
[10, 5]
[22, 18]
```

L2-Q2. (M) Peak element

P

CODE

```
def peak(a):
    a.sort()
    print(a[len(a)-2])
a=list(map(int,input().split()))
peak(a)
```

INPUT

```
1 2 3 1
```

OUTPUT

```
2
```

L2-Q3. (M) Palindromic

P

CODE

```
str=input()
a=0
c=0
for i in range(len(str)-1):
    for j in range(i+1,len(str)):
        if str[i]==str[j]:
            a=i
            c=j+1
print(str[a:c])
```

INPUT

```
babad
```

OUTPUT

```
aba
```

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L2-Q4. (M) Perfect Square

P

CODE

```
n=int(input())
count=0
while n>0:
    i=1
    while(i+1)*(i+1)≤n:
        i+=1
        n=n-(i*i)
        count=count+1
print(count)
```

INPUT

12

OUTPUT

3

Student 24 **PILLARI RAMYA** 192319027RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Search

P

CODE

```
matrix=[
    [1,4,7,11,15],
    [2,5,8,12,19],
    [3,6,9,16,22],
    [10,13,14,17,24],
    [18,21,23,26,30]
]
target = int(input())
found = False
for row in matrix:
    if target in row:
        found = True
        break
print(str(found).lower())
```

INPUT

20

OUTPUT

false

L2-Q2. (M) Target

P

CODE

```
a = list(map(int,input().split()))
target = a[-1]
numbers = a[:-1]
i = 0
j = len(numbers)-1

while i < j:
    s = number[i]+numbers[j]
    if s == target:
        print(i +1 , i+1)
        break
    elif s < target:
        i +=1
    else:
        j +=1
```

INPUT

2 7 11 15 9

OUTPUT

```
File "C:\Windows\TEMP\sandbox_192319027RC_6a7d4faf157225.24918448\main.py", line 4
    i = 0
IndentationError: unexpected indent
```

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L2-Q3. (M) Valid String

P

CODE

```
s = input()
st = []
for x in s:
    if x in "({":
        st.append(x)
    elif not st or "({".index(st.pop()) != ")".index(x):
        print("False")
        break
    else:
        print("True" if not st else "False")
```

INPUT

()[]

OUTPUT

True
True**L2-Q4. (M) Perfect Square**

P

CODE

```
n = int(input())
dp = [0]+[n]*n
for i in range(1,n+1):
    j = 1
    while j*j ≤ i:
        dp[i] = min(dp[i], dp[i - j*j]+1)
        j += 1
print(dp[n])
```

INPUT

12

OUTPUT

3

Student 25 **POTHIREDDY ROHINI** 192420054RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Search**

P

CODE

```

m=[[1,4,7,11,15],[2,5,8,12,19],[3,6,9,16,22],[10,13,14,17,24],[18,21,23,26,30]]
t=int(input())
found=False
for row in m:
    if t in row:
        found=True
        break
print("true"if found else"False")

```

INPUT

5

OUTPUT

true

L2-Q2. (M) Intersection

P

CODE

```

a=list(map(int,input().split()))
b=list(map(int,input().split()))
c=[]
for x in a:
    if x in b and x not in c:
        c.append(x)
print(c)

```

INPUT

1 2 2 1
2 2

OUTPUT

[2]

L2-Q3. (M) Valid String

P

CODE

```

s=input()
st=[]
for x in s:
    if x in "[{":
        st.append(x)
    elif not st or "([{"[st.index(st.pop())!="([{"[st.index(x):
        print("false")
        break
    else:
        print("true"if not st else"false")

```

INPUT

()[]

OUTPUT

```

Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192420054RC_6a7d47dab36799.27056831\main.py", line 6, in <module>
    elif not st or "([{"[st.index(st.pop())!="([{"[st.index(x):
                                ^^^^^^^^^^^^^^^^^
ValueError: substring not found

```


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L2-Q4. (M) Lucky

P

CODE

```
def rSum(n):
    if n==0:
        return 0
    return n%10+rSum(n//10)
def rpro(n):
    if n<10:
        return n
    return (n%10)*rpro(n//10)
n=int(input())
s=rSum(n)
p=rpro(n)
if s==p:
    print("yes")
else:
    print("no")
```

INPUT

1412

OUTPUT

yes

Student 26 **Risaph S** 192519096RC

3 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Sort

P

CODE

```
a = input()
a = a.replace("{", "")
a = a.replace("}", "")
a = set(map(int, a.split(",")))
b = sorted(a)
middle = len(b)//2
print("output= ",b)
print("Middle element:",b[2])
```

INPUT

{90, -90, 80, 66, 78, 10}

OUTPUT

```
output=  [-90, 10, 66, 78, 80, 90]
Middle element: 66
```

L2-Q2. (M) Search

— Not Submitted —

L2-Q3. (M) Isomorphic

P

CODE

```
def is_isomorphic(s,t):
    if len(s) != len(t):
        return False
    d={}
    for i in range(len(s)):
        if s[i] in d:
            if d[s[i]] != t[i]:
                return False
            else:
                d[s[i]]=t[i]
        return True
    t = "add"
    s = "egg"

    if is_isomorphic(s,t):
        print("True")
    else:
        print("False")
```

OUTPUT

True

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L2-Q4. (M) Lucky

P

CODE

```
def is_lucky(n,i):
    if i>n:
        return True
    if n % i == 0:
        return False
    return is_lucky(n-(n//i), i+1)
n = float(input())
if is_lucky(n,2):
    print("lucky")
else:
    print("Not lucky")
```

INPUT

420

OUTPUT

Not lucky

Student 27 **Sachin S** 192525045RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Repeated

P

CODE

```
nums=input()
nums=nums.split("=")[1]
nums=nums.strip("[]")
nums=list(map(int,nums.split(",")))
for i in nums:
    if nums.count(i)>1:
        print(i)
        break
```

INPUT

nums=[1,3,4,2,2]

OUTPUT

2

L2-Q2. (M) Peak element

P

CODE

```
num=list(map(int, input().strip("[]").split(",")))
for i in range(len(num)):
    if i ==0 and num[i]>num[i+1]:
        print(i)
        break
    elif i==len(num)-1 and num[i]>num[i-1]:
        print(i)
        break
    elif i>0 and i<len(num)-1:
        if num[i]>num[i-1]and num[i]>num[i+1]:
            print(i)
            break
```

INPUT

[1,2,1,3,5,6,4]

OUTPUT

1

L2-Q3. (M) Valid String

P

CODE

```
s=input()
stack=[]
for ch in s:
    if ch=="(" or ch=="{" or ch=="[":
        stack.append(ch)
    else:
        if len(stack)==0:
            print("false")
            break
        top = stack.pop()
        if (ch==")" and top != "(") or (ch=="]" and top != "[") or (ch=="}" and top != "{"):
            print("false")
            break
        else:
            if len(stack)==0:
                print("true")
            else:
                print("false")
```

INPUT

OUTPUT

L2-Q4. (M) Perfect Square

P

CODE

```
n= int(input(""))
sum=[0]+[n]*n
for i in range(1, n+1):
    j=1
    while j*j≤i:
        sum[i]=min(sum[i], sum[i-j*j]+1)
        j=j+1
print(sum[n])
```

INPUT

OUTPUT

Student 28 **SAFRIN RITHA AM** 192313059RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Starting-Ending

P

CODE

```
arr = [5,7,7,8,8,10]
target = 8
start = -1
end = -1
s=[]
for i in range(len(arr)):
    if arr[i] == target:
        if start == -1:
            start = i
        else:
            end = i
            s.append(start)
            s.append(end)
            print(s)
```

INPUT

```
[5,7,7,8,8,10]
8
```

OUTPUT

```
[3, 4]
```

L2-Q2. (M) Peak element

P

CODE

```
a = [1,2,3,1]
for i in range(1,len(a)-1):
    if a[i] > a[i+1] and a[i] > a[i-1]:
        b = a[i]
        for x in a:
            if a[i] == b:
                print(i)
                break
```

OUTPUT

```
2
```

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L2-Q3. (M) Valid String

P

CODE

```
s = "()[]{}"
stack=[]
for ch in s:
    if ch in "({":
        stack.append(ch)
    elif ch in ")}":
        if not stack or "({".index(stack.pop())!=ch.index(ch):
            print("false")
            break
    else:
        print("true" if not stack else "false")
```

OUTPUT

```
true
true
true
```

L2-Q4. (M) Perfect Square

P

CODE

```
n = 12
dp = [0] + [float('inf')]*n
for i in range(1,n+1):
    j = 1
    while j * j ≤ i:
        dp[i] = min(dp[i],dp[i - j*j]+1)
        j += 1
print(dp[n])
```

OUTPUT

3

Student 29 **Shaik Rehan** 192525110RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Sort

P

CODE

```
arr = list(map(int,input().split()))
arr.sort()
mid = len(arr)//3
print(list(arr))
print("middle element:",arr[mid])
```

INPUT

90 -90 80 66 78 10

OUTPUT

```
[-90, 10, 66, 78, 80, 90]
middle element: 66
```

L2-Q2. (M) Frequency

P

CODE

```
arr = list(map(int,input().split()))
max_count = 0
element = None
for x in arr:
    count = arr.count(x)
    if count>max_count:
        max_count = count
        element = x
        print(element)
        print(max_count)
```

INPUT

9 10 8 3 4 5 9 8

OUTPUT

```
9
2
```

L2-Q3. (M) Haystack

P

CODE

```
n = list(map(int))
```

OUTPUT

```
-1
explanation: letto did not occur in leetcode,so we return -1
```

L2-Q4. (M) Perfect Square

P

CODE

```
n = int(input())
count = 0
while n>0:
    x = int(n**0.5)
    n -= x*x
    count +=1
print(count)
```

INPUT

1

OUTPUT

1

Student 30 **SIVACHANDRABOSE S** 192319030RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Frequency**

P

CODE

```
arr=[1,2,8,3,2,2,2,5,1]
freq={}
for x in arr:
    freq[x]=freq.get(x,0)+1
print("Frequency:", freq)
element=max(freq, key=freq.get)
print("Highest frequency:", freq[element])
print("Element:", element)
```

OUTPUT

```
Frequency: {1: 2, 2: 4, 8: 1, 3: 1, 5: 1}
Highest frequency: 4
Element: 2
```

L2-Q2. (M) Repeated

P

CODE

```
num=[3,1,3,4,2]
for i in range(len(num)):
    if num.count(num[i])>1:
        print("Repeated number:", num[i])
        break
```

OUTPUT

```
Repeated number: 3
```

L2-Q3. (M) Haystack

P

CODE

```
needle=("Needle: ")
haystack=("Haystack: ")
print(haystack.find(needle))
```

INPUT

```
haystack= "sadbutsad", needle="sad"
```

OUTPUT

```
-1
```

L2-Q4. (M) Perfect Square

P

CODE

```
n=int(input("Enter the number: "))
for i in range(n,1):
    if n%i:
        print("Perfect square")
        break
    else:
        print("Not a perfect square")
```

INPUT

```
13
```

OUTPUT

```
Enter the number:
```

Student 31 **Smohit Arman Swain** 192572092RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Repeated

P

CODE

```
A= [3,1,3,4,2]
for i in range (len(A)):
    for j in range (i+1,len(A)):
        if A[i]==A[j]:
            print(A[i])
            break
```

OUTPUT

3

L2-Q2. (M) 2D Multiplication

P

CODE

```
A=[[1, 2],
   [5, 3]]
B=[[2, 3],
   [4, 1]]
C=[[0,0],
   [0,0]]
for i in range(2):
    for j in range(2):
        C[i][j]=0
        for k in range(2):
            C[i][j]= C[i][j]+ A[i][k] * B[k][j]
print("matrix multiplication result:")

for i in range(2):
    for j in range(2):
        print(C[i][j],end=" ")
    print()
```

OUTPUT

```
matrix multiplication result:
105
2218
```

L2-Q3. (M) Number of Ups, Lows

P

CODE

```
str1=[1,7,6,9,5]
```

OUTPUT

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L2-Q4. (M) Lucky

P

CODE

```
n = int(int())

if n%2==1:
    print("no")
else:
    print("yes")
```

INPUT

1412

OUTPUT

yes

Student 32 **SUNDARA VARADHAN K S** 192319033RC

3 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Search

— Not Submitted —

L2-Q2. (M) Intersection

P

CODE

```
def intersect(nums1 = [1,2,2,1], nums2 =[2,2]):
    d, res = {}, []
    for x in nums1:
        d[x]=d.get(x) +1
        if d.get(x,0) > 0:
            res.append(x)
            d[x] -= 1
    return res
print(intersect(nums1, nums2))
```

INPUT

nums1 = [1,2,2,1], nums2 = [2,2]

OUTPUT

L2-Q3. (M) Isomorphic

P

CODE

```
a = "egg"
b = "add"
d = {}
for x,y in zip(a,b):
    if x in d and d[x]!=y:
        print("false")
        break
    d[x] = y
else:
    print("true")
```

OUTPUT

```
true
true
true
```

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L2-Q4. (M) Lucky

P

CODE

```
n = 1412
s = 0
p = 1
for i in str(n):
    s +=int(n)
    p*=int(n)
if s==p:
    print("lucky")
else:
    print("not lucky")
```

INPUT

1412

OUTPUT

not lucky

Student 33 **Thanikonda Aditya** 192565068RC

4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Minimum

P

CODE

```
n=[4,5,6,7,0,1,4]
n.sort()
print(n)
```

OUTPUT

```
[0, 1, 4, 4, 5, 6, 7]
```

L2-Q2. (M) Peak element

P

CODE

```
def peak(a):
    a.sort()
    print(a[len(a)-2])
a=list(map(int,input().split()))
peak(a)
```

INPUT

```
1 2 1 3 5 6 4
```

OUTPUT

```
5
```

L2-Q3. (M) Isomorphic

P

CODE

```
s1=input()
s2=input()
d={}
for i in range(len(s1)):
    if s1[i] in d:
        if d[s1[i]]!=s2[i]:
            print("false")
            break
    else:
        d[s1[i]]=s2[i]
else:
    print("true")
```

INPUT

```
egg
add
```

OUTPUT

```
true
```

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L2-Q4. (M) Perfect Square

P

CODE

```
n=int(input())
count=0
while n>0:
    i=1
    while(i+1)*(i+1)≤n:
        i+=1
        n=n-(i*i)
        count+=1
print(count)
```

INPUT

12

OUTPUT

3

Student 34 **Thumati Sai Praneeth** 192525434RC

4 / 4 submitted

Level 2 — 5 marks/question**L2-Q1. (M) Frequency**

P

CODE

```
a = [1,2,8,3,2,2,2,5,1]
for i in set(a):
    print(i,a.count(i))
print("highest frequency:",max(a.count(i) for i in set(a)))
```

OUTPUT

```
1 2
2 4
3 1
5 1
8 1
highest frequency: 4
```

L2-Q2. (M) 2D Multiplication

P

CODE

```
a = [[1,2],[5,3]]
b = [[2,3],[4,1]]
c = [[0,0],[0,0]]
for i in range(2):
    for j in range(2):
        for k in range(2):
            c[i][j]+=a[i][k]*b[k][j]
print(c)
```

OUTPUT

```
[[10, 5], [22, 18]]
```

L2-Q3. (M) Haystack

P

CODE

```
haystack = "sadbussad"
needle = "sad"
print(haystack.find(needle))
```

OUTPUT

```
0
```

L2-Q4. (M) Perfect Square

P

CODE

```
n = 12
count = 0
while n>0:
    i = 1
    while(i+1)*(i+1)≤n:
        i+=1
        n=n-(i*i)
        count+=1
print(count)
```

OUTPUT

```
3
```